

Yene Irvine

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TECHNICAL SKILLS

Data Engineering: SQL, Data warehousing (Snowflake, DB2), ETL / ELT pipelines
Backend: Node.js (Express, NestJS), FastAPI, Java, PostgreSQL, SQLAlchemy, REST & SOAP APIs
Frontend: TypeScript, JavaScript (React), Tailwind CSS, Redux, Webpack, Babel
Cloud & DevOps: AWS, Azure, GCP, Vercel, Docker, Terraform, Redis, GitHub Actions, RabbitMQ, CI/CD pipelines
Machine Learning: PyTorch, scikit-learn, Pandas, NumPy, matplotlib, Weights and Biases (WandB), HPC compute

EXPERIENCE

Data Engineer

Sep. 2024 – Present

Fuel Transport Inc.

Montreal, QC

- Built and integrated a Snowflake Rate Editor (FastAPI, React) to enable real-time editing of client rates, powering 100s of daily API calls and contributing to \$1.5M+ in revenue over 5 months.
- Developed and implemented a custom Azure Data Factory pipeline to migrate data from IBM DB2 to Snowflake, replacing Fivetran and saving \$40K+ annually in third-party costs.
- Optimized CI/CD pipelines (GitHub Actions, Docker, Azure Container Apps) to streamline deployment of the Snowflake Rate Editor's frontend and backend containers, improving reliability and reducing overhead.
- Leveraged Azure Monitor, Application Insights, and related Azure services to track API performance, analyze traffic patterns, and proactively resolve issues, ensuring high availability for enterprise clients.

Controls Engineer

Apr. 2019 – Jan. 2022

MCA Consultants

Charlottetown, PE & Halifax, NS

- Developed and commissioned 25 control narratives for utilities and HVAC control systems, integrating PLCs and process equipment, and working with contractors to enhance the Building Automation System.
- Created a program in Excel to determine relieving requirements for pressurized vessels based on NFPA and ASME standards; used in designing venting configurations for 50+ major vessels.
- Led a team to commission all process, utility, and HVAC systems and associated equipment under budget.

SIDE PROJECTS

ReliefGuard (reliefguard.ca) | *Next.js, React, TailwindCSS, Vercel*

- Developed and deployed a web application for pressure relief valve sizing calculations, replacing legacy VBA Excel workbooks used in industrial engineering. Built with Next.js, TypeScript, and TailwindCSS, implementing NFPA 30, API 521, and ASME VIII compliance standards.
- Architected real-time calculation engine with React Context state management, featuring complex heat transfer formulas, gas flow calculations, and automated validation. Deployed on Vercel with CI/CD.
- Modernized engineering workflows by reducing manual error risk through automated validation, providing formula documentation, and enabling real-time design basis flow calculations across multiple vessel failure scenarios.

Detecting Breathing Irregularities with Deep Learning | *3D CNN, U-Net, PyTorch, HPC, WandB*

- Developed, deployed a 3D Convolutional Neural Network with UofC Biometrics Lab to detect sleep-related medical emergencies via thermal video, leveraging WandB for real-time tracking at Calgary Drop-In Centre.

AI DJ Setlister | *Flask, Python, Spotify Web API, OAuth2, Genetic Algorithms, AWS EC2*

- Built a cloud-hosted Flask app allowing users to log in via Spotify OAuth2, select or paste playlists, and generate optimized DJ setlists using algorithms that consider key compatibility, energy progression, and beat matching

EDUCATION

University of Calgary

Calgary, AB

M.Eng., Software Engineering (GPA 3.94)

Graduated May 2024

Dalhousie University

Halifax, NS

B.Eng., Chemical Engineering (GPA 3.80)

Graduated Apr. 2018

CERTIFICATIONS

AWS Certified Cloud Practitioner CLF-C02 (2024)

Microsoft Azure Fundamentals AZ-900 (2024)